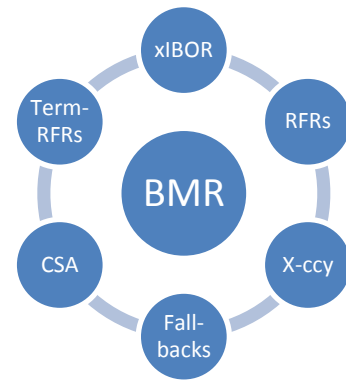




Benchmark Reform: Hedging, and impacts on legacy contracts

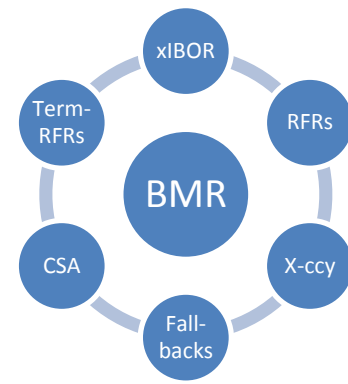
Paul Brooks – Head of Traded Market Risk
Bank of Ireland

Background: Organisation



- Exposure – activity
 - Corporate and retail
 - Balance sheet hedging
 - Customer derivatives execution
 - Discretionary trading
- Exposure – products
 - Euribor, Eonia, £LIBOR, SONIA, cross-market;
 - Derivatives, loans, CSAs, indirect exposure;
- Derivatives books:
 - 60% cleared
 - 35% CSA bilateral
 - <5% uncollateralised

Overview



Challenges I: Final state of reform

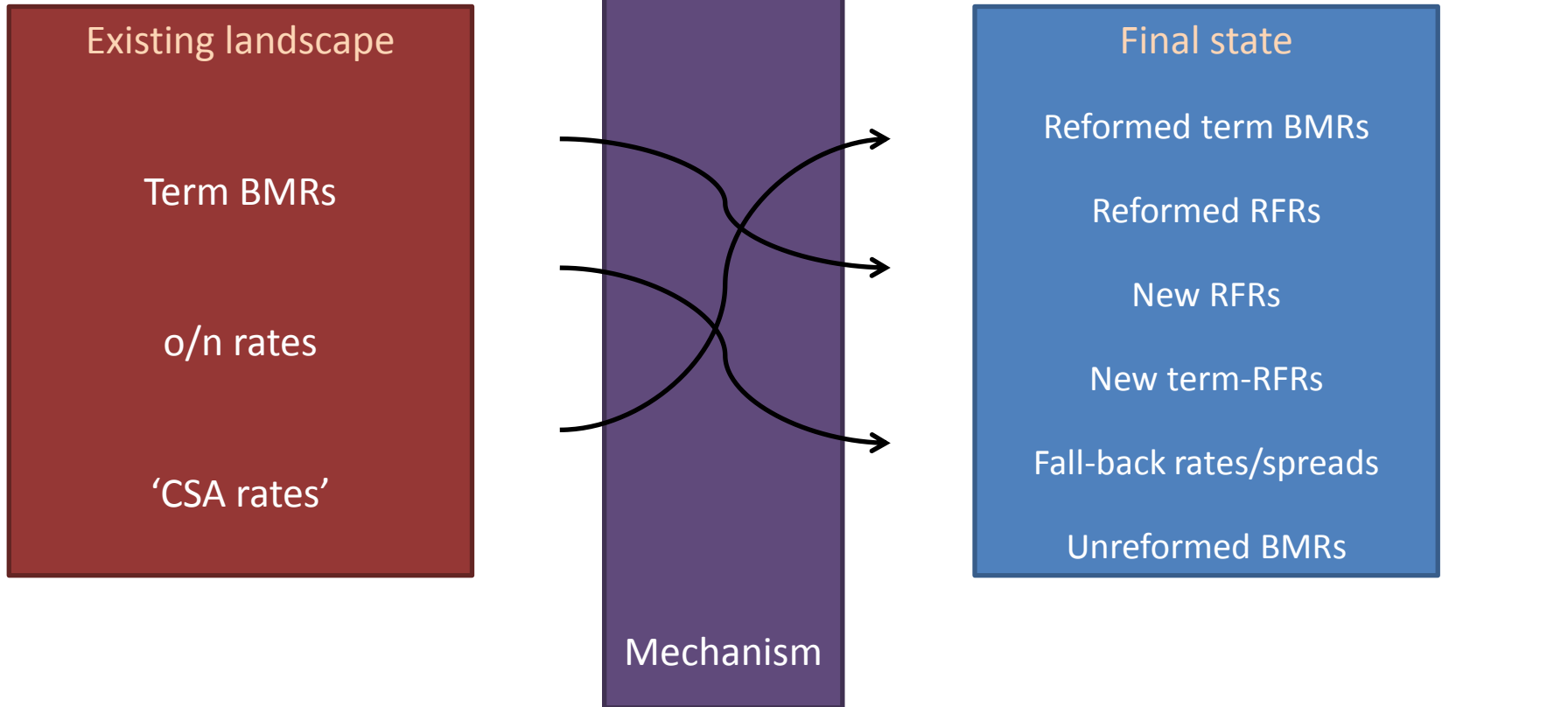
Challenges II: Transition mechanism

Challenges III: Current BMR landscape

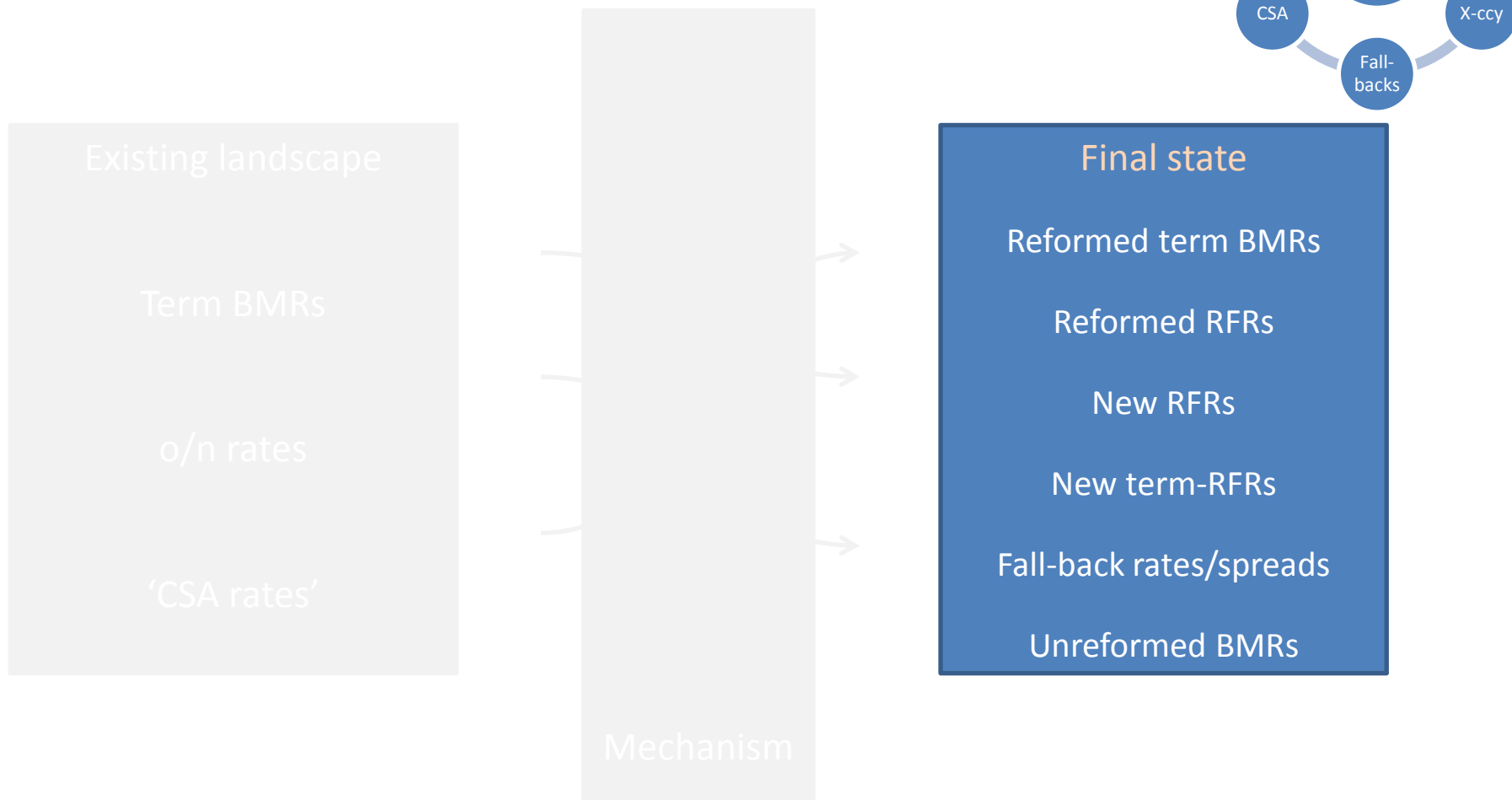
Hedging scenarios

Summary and conclusion

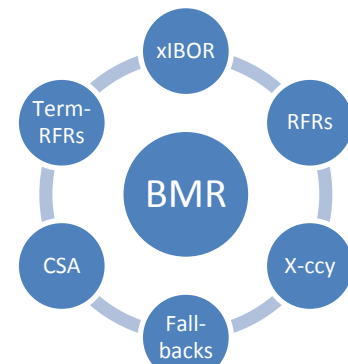
Existing landscape vs final state



Challenges I: Final state uncertain



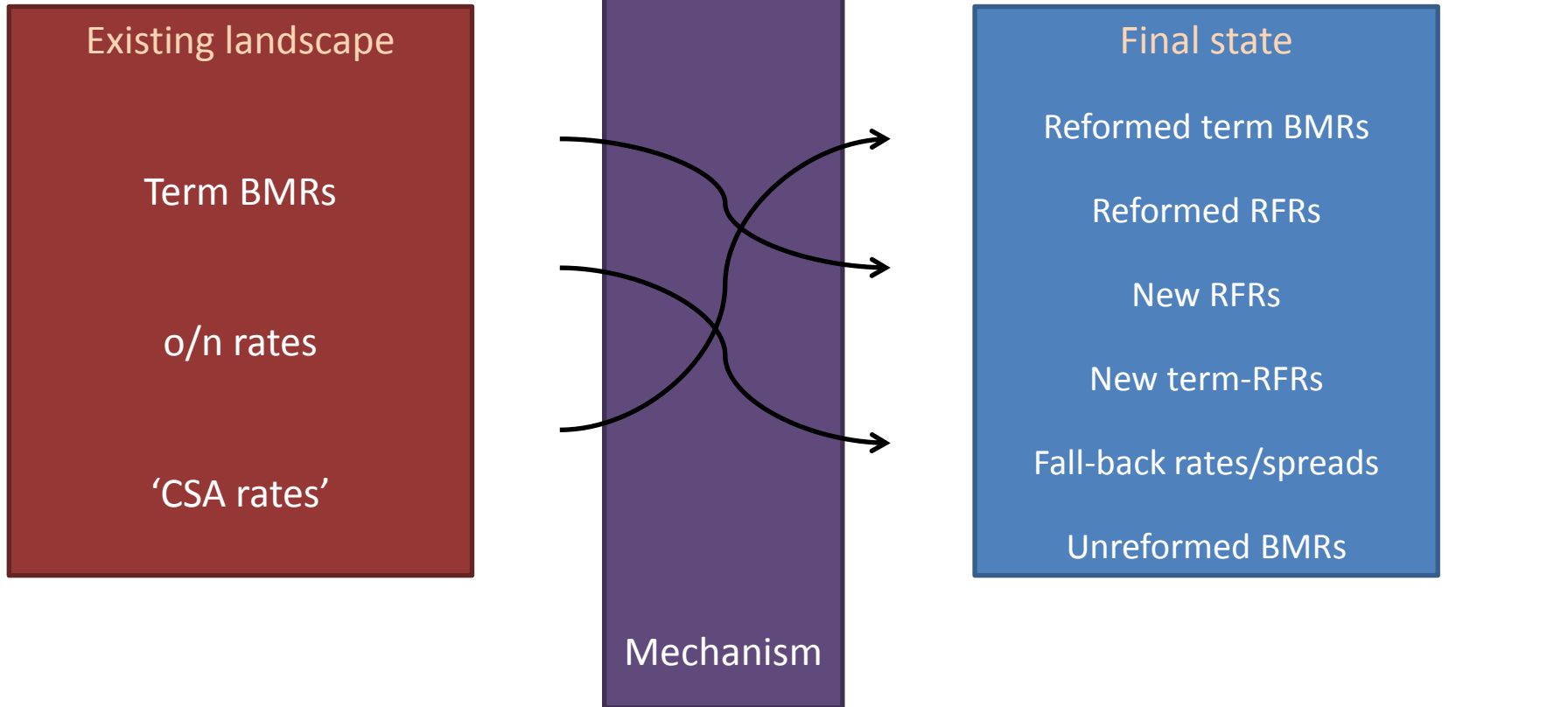
Challenges I: Final state uncertain



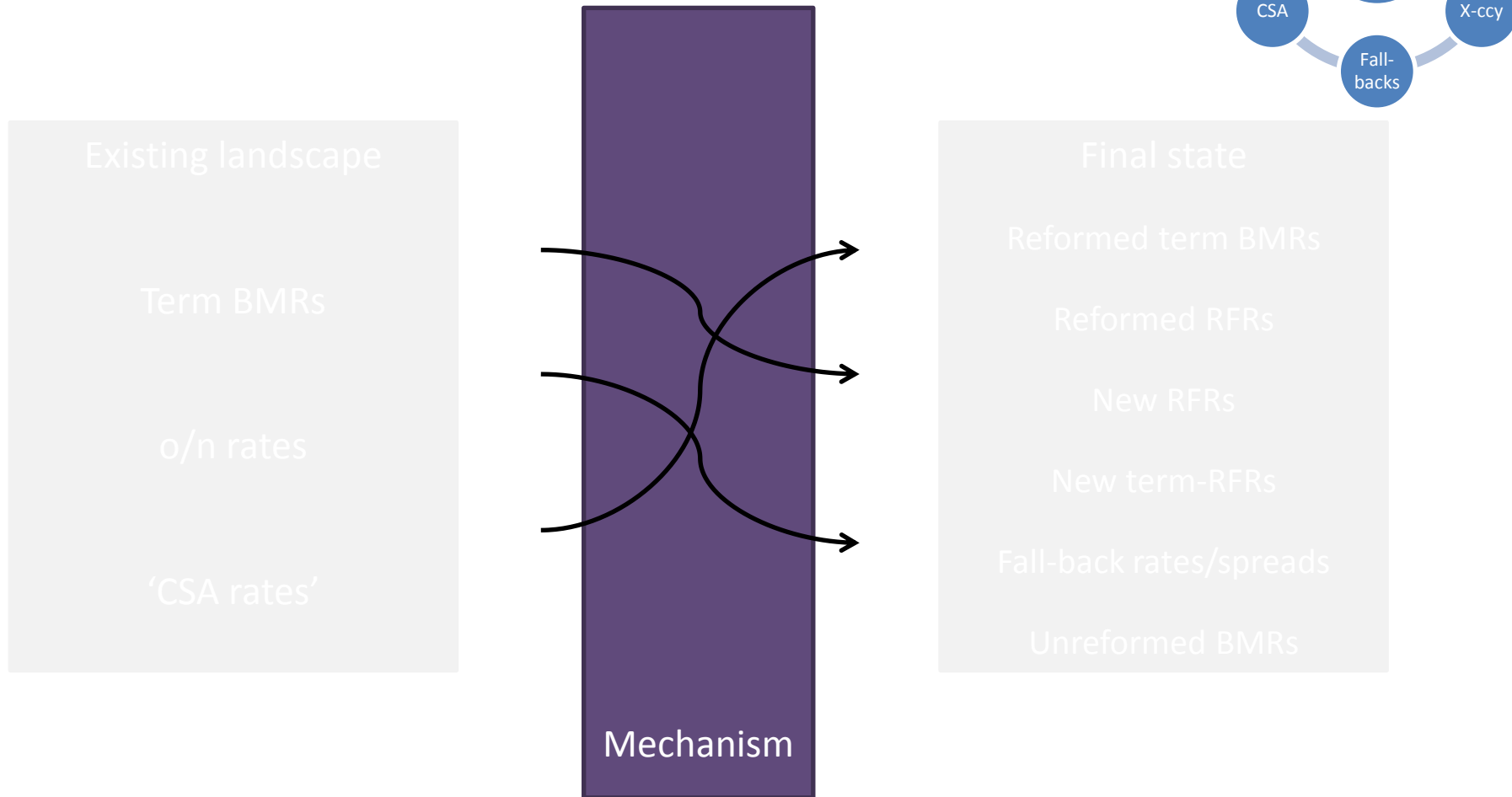
Ccy	Term BMR	o/n BMR	CSA rates
EUR	Euribor ↓ Hybrid Euribor? ↓ Term €str?	Eonia ↓ Ester (€str)	Eonia ↓ ?
GBP	LIBOR ↓ Term rSonia?	Sonia ↓ rSonia	Sonia ↓ rSonia
USD	LIBOR ↓ Term SOFR	FF ↓ SOFR	Sonia ↓ SOFR?

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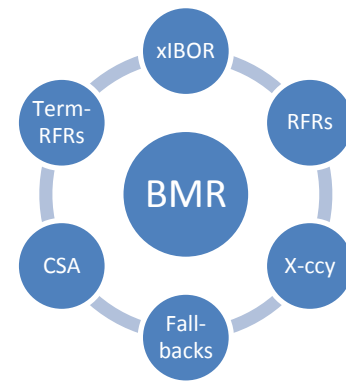
Existing landscape vs final state



Challenges II: Multiple migration mechanisms

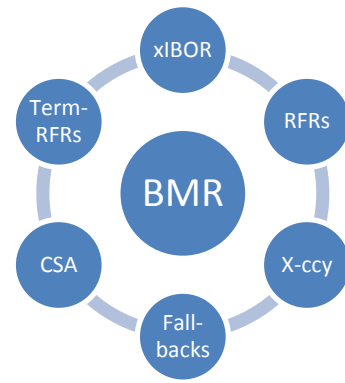


Challenges II: Multiple migration mechanisms



Mechanism	Notes
Direct replacement	E.g. reformed Sonia
Recalibrated replacement	New rate replaces old rate, but with recalibration to ensure minimal value transfers e.g. Eonia → €str
Un-managed transition	Market left to manage without recalibration or transition
Negotiated transition	Migration to new rates negotiated bilaterally e.g Eonia
Transition to existing rate	Dynamics of migration known, hedgeable e.g. £LIBOR
Transition to new rate	Dynamics of transition unknown, unhedgeable e.g. €str
Fall-backs	Proscribed mechanisms, no control
No clear single successor rate	CSA rates?

Challenges II: Multiple migration mechanisms



Simplicity

✗ Direct replacement

✗ Recalibrated replacement

✗ Transition to existing rate

Impact

✗ Un-managed transition

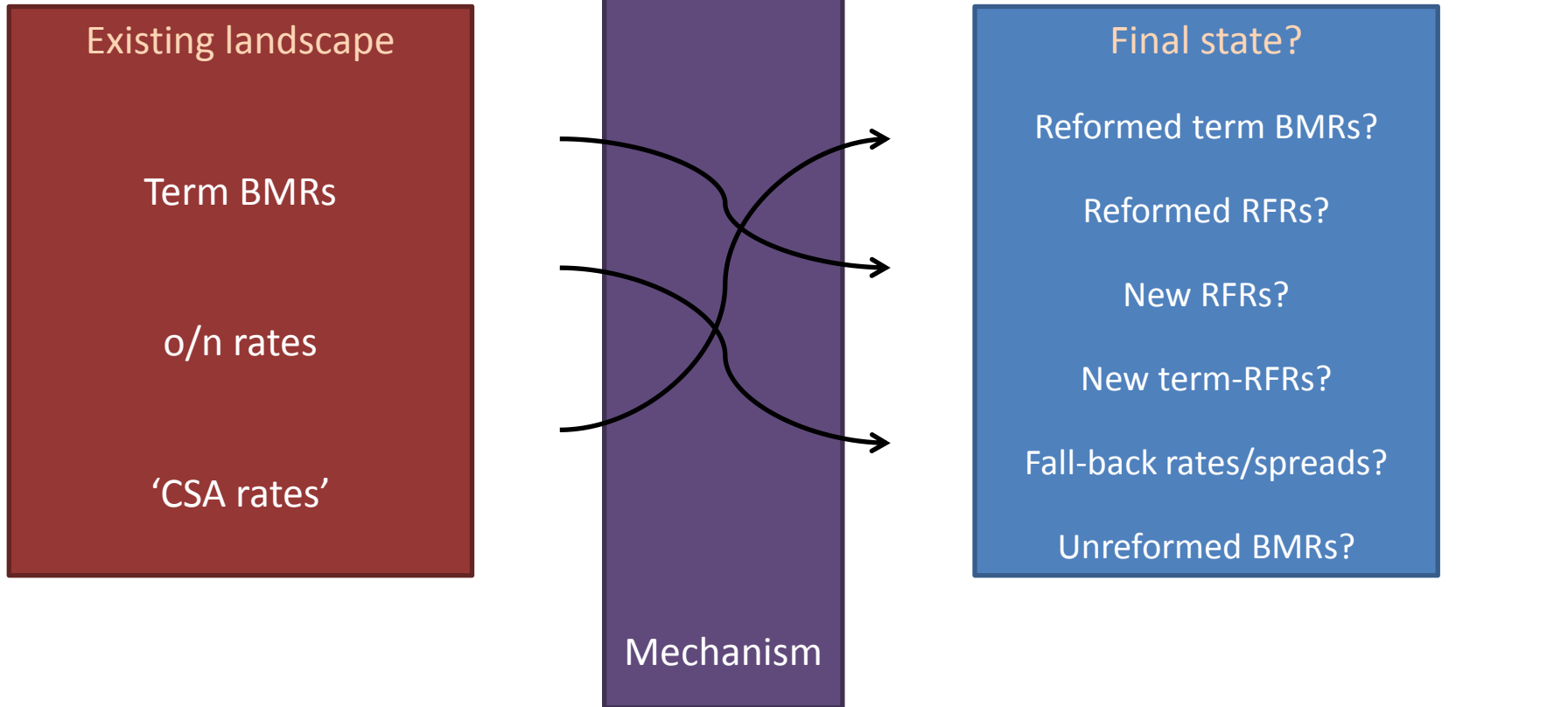
✗ Fall-backs

✗ Renegotiated replacement

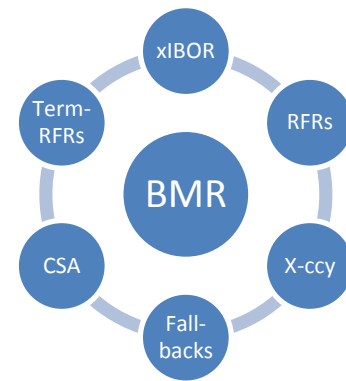
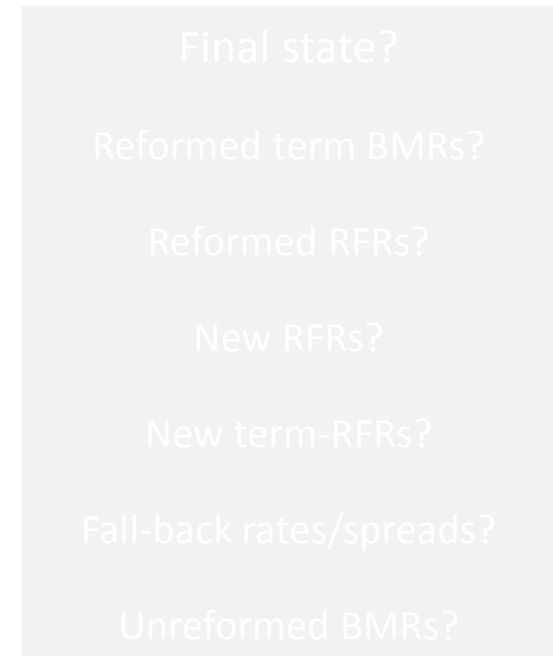
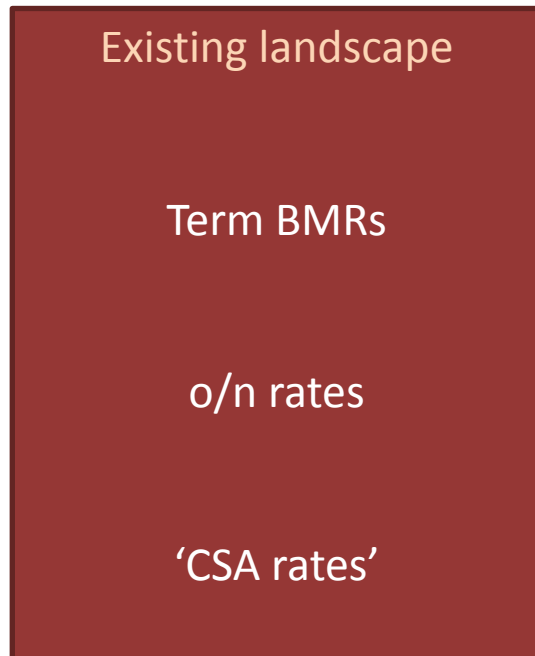
✗ Transition to new rate

✗ No clear successor rate

Existing landscape vs final state

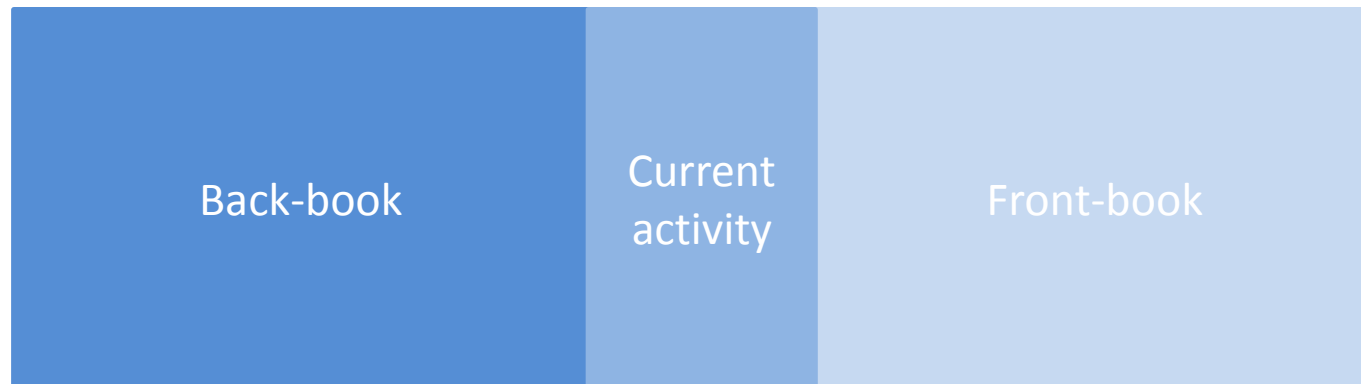
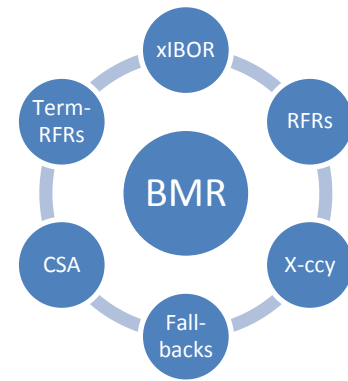


Challenges III :Diversity of exposure



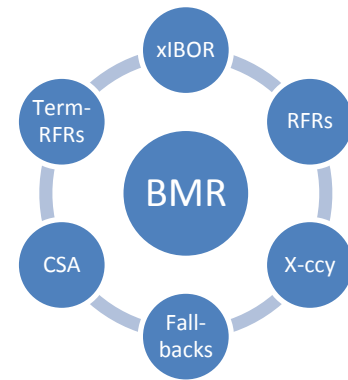
Challenges III :Diversity of exposure

- Markets referencing BMRs are an ecosystem, dependent on multiple other participants, and transition is constrained by this
 - Interbank/CCP
 - Non-bank (derivatives, loans, indirect exposure)
 - Internal exposures (treasury, intra-group)
 - Products: cash, derivatives, bonds, issuance etc.
- Approach segmented across product classes
 - E.g. loans vs derivatives
- Approach segmented within institutions
 - Back-book, 'now-book', front-book



Challenges IV: Other dependencies & uncertainties

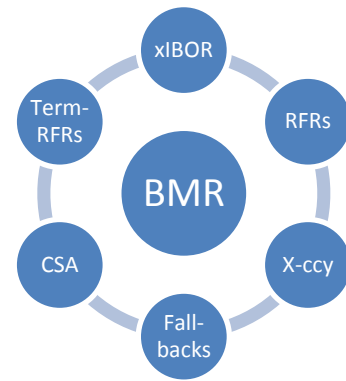
- Development of derivative markets in new BMRs – SOFR market is first mover, slow progress
- Timelines: growing certainty, but not fully there
- CCP support – chicken/egg – SOFR good proof of concept
- Fall-back protocols: complex spread methodologies; not yet finalised; fragmented approach
- Hedge accounting treatment
- Bilateral negotiation (bilateral margining repapering++) – capturing value transfers may be a challenge
- Possible intermediate steps before final transition



Preparation constrained by lack of certainty on many fronts

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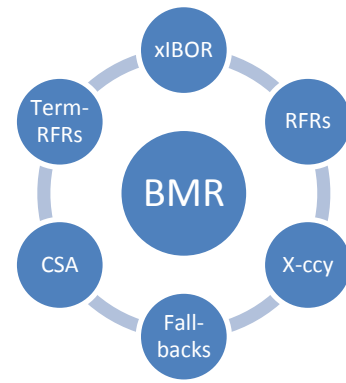
Impacts and strategy



- Impacts
 - Value transfers – may no be possible to capture all
 - Risk measurement – successor rate uncertain, risk/valuation impact unknown
 - Risk measurement – models based on decades old BMRs
 - Structural positioning
 - Documentation
 - Re-papering CSAs (a la VM margining regulations)
- Strategy
 - Identifying inventory
 - Manage front book
 - Hedging strategy: pre-hedge, macro-hedge, micro-hedge?
 - Timing likely to be driven by mixture of regulators, CCPs and market makers

Inventory → Manage front-book → Pre-hedge → Macro-hedge → Micro-hedge

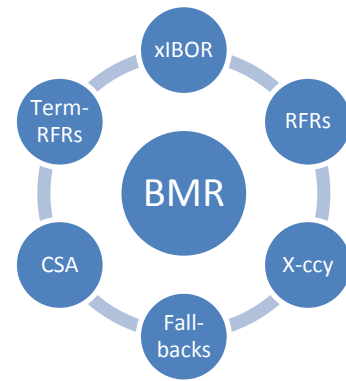
Hedging scenarios



- Direct replacement with new rate
 - E.g. Sonia → rSonia
 - Possibility for Euribor → hEuribor
 - Challenge to pre-hedge, unlikely to be dual-derivatives market
 - Valuation impact transparent, but likely to be material and largely unhedgeable

- Direct replacement with existing rate
 - Possibility for £ LIBOR → rSonia
 - Hedging potentially possible now, liquidity issues with long dated exposure
 - Convexity effects require modelling
 - Possible to capture value transfers

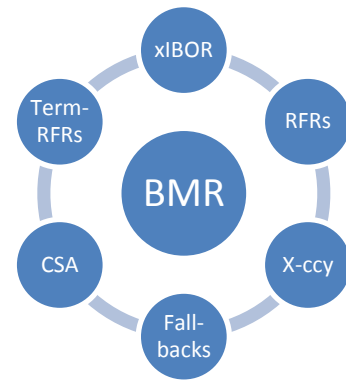
Hedging scenarios



- Recalibrated replacement
 - Transition to new rate -/+ spread
 - Possibility for Eonia → €str
 - Also a feature of some LIBOR fall-backs
 - Can minimise valuation impact, depending on mechanism
 - Hedging a challenge, but can in theory position for transition

- Negotiated transition
 - Possibility for Eonia → €str
 - Requires dual market to be present
 - Hedge-able when new derivatives market emerges
 - Valuation impact transparent, but could be challenge to capture

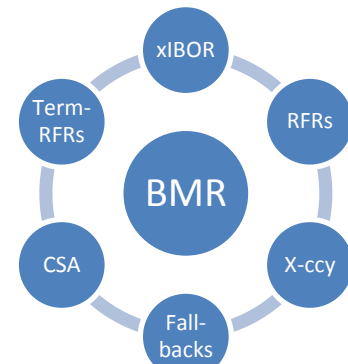
Hedging scenarios



- CSA rates
 - No regulatory pressure in some situations (e.g. Eonia)
 - Reliance on defunct rates a risk
 - Likely requires re-papering exercise a la bilateral margining regs
 - Challenge for industry, as value transfers likely
 - Hedging exposure possible, but reliance on Eonia markets

- Fall-back
 - Mechanisms generally relate to new RFRs
 - Hedgeable to the extent the derivatives markets in new RFRs exist
 - Fragmented approach across jurisdictions
 - Convexity and rate timing effects

Final state of reform

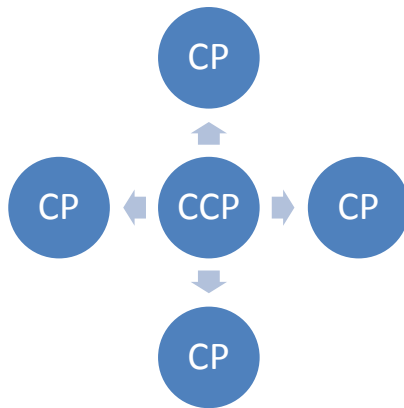
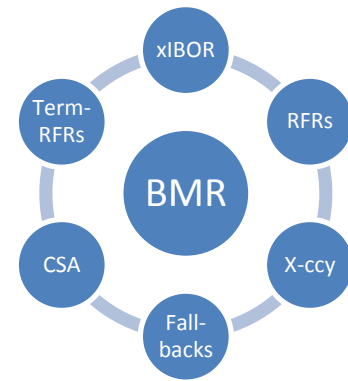


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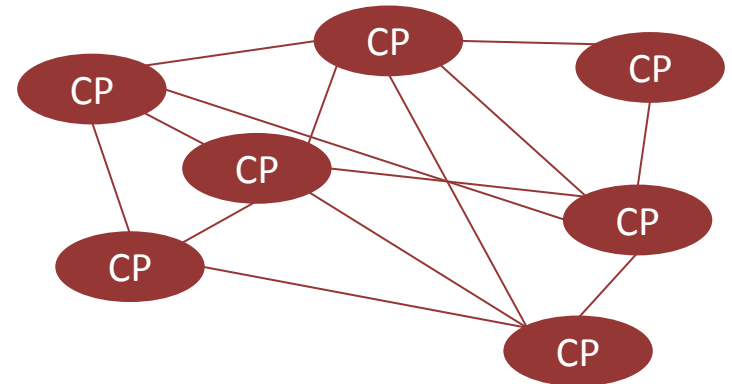
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Other considerations

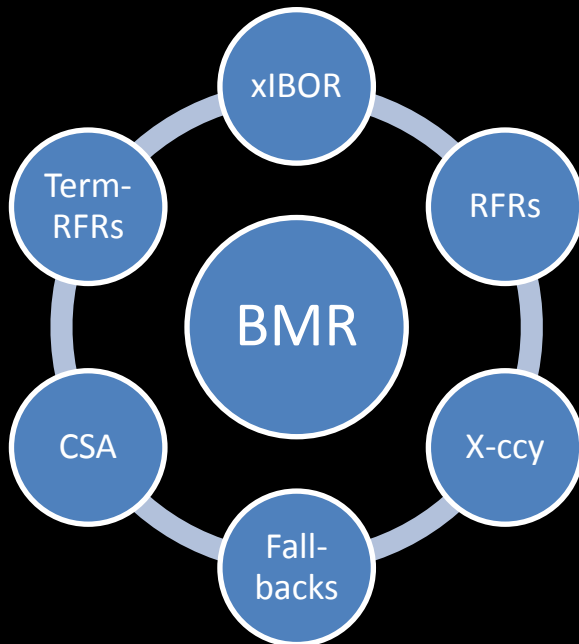
- Cross market: Cross-currency markets a particular challenge: un-cleared, timelines, transition mechanisms, and nature of successor rates differs for each 'leg'
- Internal pricing: FTP and other internal pricing structures may dictate hedging strategy
- CCP managed or bilateral negotiation? Centralisation of problem versus control over process



vs



Summary

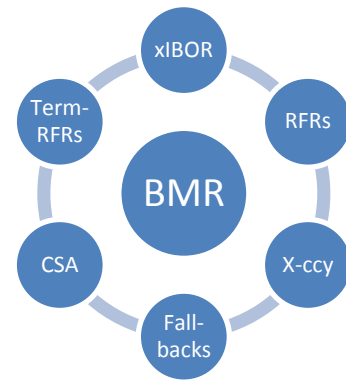


- Hedging strategy dependent on
 - Timelines
 - Successor rate
 - Transition mechanism
 - Development of derivatives markets
- Not enough certainty at present to hedge material exposures
- Gradually gaining certainty over final state (60% there), but implementation will be staggered
- Hedging generally only possible where new/old rates run concurrently
- Cross-market products an additional challenge

Useful links

ECB round table on euro risk-free rates:

https://www.ecb.europa.eu/paym/intro/events/html/20181109_euro_risk-free_rates.en.html



Title

11.55 What will firms need to do to rehedge underlying exposures in the wake of rates reform?

- What approaches will firms look to take when rehedging the back book
- How to handle the back book after the demise of Libor and Eonia
- The timing of the implementation:

